

Almost-Idempotent Stochastic Maps — internal lab-book

(scientific-exploration repo; see PRD.md / CLAUDE.md)

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1 Roadmap and rigour status

This lab-book tracks the report-facing slice of the registry in `argument/`. The north star is `op-classical`; its status is *open*, and the status-ledger anchor is 1. The definitions used by the registry are the canonical shards `def-stochastic`, `def-almost-idempotent`, `def-near-positive-projection`, `def-signed-idempotent`, `def-negative-mass`, `def-height`, `def-visible-set`, `def-invisible-mass`, and `def-exposed`. This report references those identifiers; it does not introduce parallel definitions.

The rigorous line is deliberately narrow. A result may be called rigorous here only when it is byte-matched to a local theorem under `refs/`, validated in an `af` workspace, or Lean-proved. At the time of this shard, the only registry results reproduced as rigorous are the seven `af`-validated entries 1, 2, 3, 4, 5, 6, and 7. Every other registry result remains anchored only in the status ledger, with its lower-rung status left intact.

The working headline is also status-sensitive. The inherited campaign distinguishes the realizable-family linear law from the worst-case `sqrt`-envelope: the linear relation is the mechanism seen in the reduction record, while the `sqrt` scale is the stability target. The sharp-exponent obstruction is `ex-hume`, anchored at 1, but it is still `proved-mod-audit` here rather than one of the rigorous results. The bridge between the signed and stochastic pictures is the validated result 1; crossing between those pictures should cite that registry id rather than re-normalising notation locally.

2 The signed-stochastic bridge

Lemma 1 (Registry contract for lem-classical-equiv). *The signed-idempotent and stochastic-idempotent formulations of classical stability are equivalent up to universal constants: Q row-stochastic with $\|Q^2 - Q\| \leq \eta$ gives $P = \text{theta}(2Q - 1)$ signed affine retraction with $\|P - Q\| \leq C \eta$ and neg mass $\delta \leq C \eta$, and conversely row-normalising p_i^+ gives Q with $\|P - Q\| \leq 2 \delta$, $\|Q^2 - Q\| \leq 6 \delta + 4 \delta^2$.*

Status. proved; af: validated. This is one of the rigorous in-repo results.

Proof tree. The registry shard records a 29-node adversarial tree with root `validated` and clean taint. The human-readable export is `proofs/lem-classical-equiv/export.md`; the export also records the earlier missing-definition challenge and its repair by importing `def-negative-mass`.

Role in the chain. This bridge is the authorized way to move between `op-classical`'s stochastic formulation and the signed exact-idempotent registry. It is a dependency for the inherited simplex, rank-one, and approximate-simplex reductions, but those downstream reductions remain `proved-mod-audit` unless and until they clear an L0 gate.

3 Height collapse

Proposition 2 (Registry contract for obs-height-collapse). *Height collapse: for an exact signed idempotent P with $0 < \delta(P) \leq 1/4$, nonempty visible set $W(P)$, and hidden top vertex v (height $H = \text{dist}_1(p_v, \text{conv } W)$ maximal among rows), the invisible mass σ_v and the row negative mass ν_v satisfy $H * (1 - \sigma_v) \leq \nu_v * (2 + 4*\delta)$.*

Status. proved; af: validated. This is one of the rigorous in-repo results.

Proof tree. The registry shard records a 19-node adversarial tree with root **validated** and taint 19/19 clean. The export is `proofs/obs-height-collapse/export.md`. Earlier compound "hence" clauses were excluded from the contract before validation; this section records only the validated inequality.

Role in the chain. This result explains why invisible mass alone does not finish the Kernel/(EX) route: height is controlled only after pricing the complementary mass. A future dimension-free cap on the relevant invisible mass would combine with this inequality, but that cap is not proved here.

4 Mass split bookkeeping

Lemma 3 (Registry contract for `lem-mass-split`). *Mass split: for an exact signed idempotent P and any row index v , writing $a_j = P_{\{vj\}}$, $a_j^{+} = \max(a_j, 0)$, $a_j^{-} = \max(-a_j, 0)$, and $nu_v = \sum_j a_j^{-}$, one has $\sum_j a_j^{+} = 1 + nu_v$.*

Status. `proved`; `af`: `validated`. This is one of the rigorous in-repo results.

Proof tree. The registry shard records a clean 9-node adversarial tree with root `validated` and taint 9/9 `clean`. The export is `proofs/lem-mass-split/export.md`.

Role in the chain. This is the mass bookkeeping dependency factored out of the first `conj-halo-collapse` elevation attempt. Its purpose is to keep later halo-collapse proof trees from re-deriving the same row-sum identity in multiple sibling branches.

5 Convex outsourcing

Lemma 4 (Registry contract for lem-residual-lower). *Convex outsourcing:* let C be the convex hull of finitely many points of $\mathbb{R}^n$, and suppose $p = \sum_{j=1}^m c_j p_j + (1 - s) q$ with points p_j, q in $\mathbb{R}^n$, coefficients $c_j \geq 0$, $s = \sum_{j=1}^m c_j < 1$, and $\text{dist}_1(p_j, C) \leq \text{dist}_1(p, C)$ for every j ; then $\text{dist}_1(p, C) \leq \text{dist}_1(q, C)$.

Status. proved; af: validated. This is one of the rigorous in-repo results.

Proof tree. The registry shard records 32 validated live nodes, 36 total nodes including 4 archived nodes, root validated, and taint 36/36 clean. The export is proofs/lem-residual-lower/export.md.

Role in the chain. This is a frame-free convex geometry dependency for conj-halo-collapse. It supplies the lower residual-distance comparison used when a hidden row is split into ordinary mass plus a residual term.

6 Residual distance bound

Lemma 5 (Registry contract for lem-residual-upper). *Residual distance bound: let C be the convex hull of finitely many points of R^n , let $b_1..b_M, c_1..c_N \geq 0$ with $m = \sum_j b_j - \sum_k c_k > 0$, let p_j, r_k be points of R^n with $q = (\sum_j b_j p_j - \sum_k c_k r_k) / m$, and let $D_k \geq 0$ satisfy $\|x - r_k\|_1 \leq D_k$ for all x in C and each k ; then $m * \text{dist}_1(q, C) \leq \sum_j b_j * \text{dist}_1(p_j, C) + \sum_k c_k * D_k$.*

Status. proved; af: validated. This is one of the rigorous in-repo results.

Proof tree. The registry shard records 49 validated live nodes, 52 total nodes including 3 archived nodes, root validated, and taint 52/52 clean. The export is proofs/lem-residual-upper/export.md. The tree size is above the 12-node brittleness target, so the tracked refactor warning remains honest rather than being hidden in prose.

Role in the chain. This is the companion frame-free convex geometry dependency for conj-halo-collapse. Together with 3 and 4, it packages the residual-recombination estimate needed by the halo-collapse bridge, now itself validated in 6.

7 Halo-robust height collapse

Lemma 6 (Registry contract for conj-halo-collapse). *Halo-robust height collapse:*
*for an exact signed idempotent P with $0 < \delta(P) \leq 1/4$, nonempty visible set $W(P)$, and hidden top vertex v of height H , let σ be the invisible mass of v , σ_g the halo-robust invisible mass (the positive coefficient mass v places on rows at $\ell-1$ distance $> \tau/4$ from $\text{conv } W$, $\tau = \sqrt{\delta(P)}$), and ν_v the row negative mass; then $H * (1 - \sigma_g) \leq (\sigma - \sigma_g) * \tau/4 + \nu_v * (2 + 4*\delta)$.*

Status. proved; af: validated. This is one of the rigorous in-repo results.

Proof tree. The registry shard records a 20-node adversarial tree, root validated, taint 20/20 clean (run 2 on the factored workspace). The export is `proofs/conj-halo-collapse/export.md`. The three af-validated dependencies 3, 4, and 5 are imported as externals, so the root rests only on validated or cited inputs.

Elevation history. Run 1 (pre-factoring) ballooned past the node cap (49 live nodes over the cap of 40) and was aborted as a structure signal, not a mathematical refutation: the mass-split book-keeping and the residual-distance estimates were being re-derived inline across siblings. Factoring those into the three registry sub-lemmas above cured it; run 2 closed at 20 nodes.

Role in the chain. This bridge refines the validated collapse bound 2 by pricing recipients inside the $\tau/4$ -halo of the visible hull at their actual distance rather than the worst case H , so the bound stays non-vacuous even when the raw invisible mass exceeds one through self or halo mass (1). Together with a halo-robust cap $\tilde{\sigma}_g \leq 1 - c$ for hidden top vertices it would give $H = O(\tau)$, the height cap of the Kernel Conjecture 1. That cap is OPEN: nothing in this section promotes the kernel chain.

8 Factorization bound

Lemma 7 (Registry contract for lem-factorization). *Factorization bound: let P be an exact signed idempotent (square real matrix with $P^2 = P$ and all row sums equal to 1), let $\delta(P) = \max_i \sum_j \max(-P_{ij}, 0)$, let $k = \text{rank}(P)$, and let $U = (u_1, \dots, u_k)$ be an actual-row basis of P (the rows $p_{\{u_1\}}, \dots, p_{\{u_k\}}$ of P form a basis of the row space of P) whose Gram volume satisfies $\text{Vol}(U) \geq (1/2) * \text{Vol}_{\max}(P)$, where $\text{Vol}_{\max}(P)$ is the maximum Gram volume over all actual-row bases of P ; define coordinates $a_t(j)$ by $p_j = \sum_t a_t(j) p_{\{u_t\}}$, and for each pivot index s in $\{1, \dots, k\}$ set $\beta_s(j) = P_{\{u_s\}j}$, $\lambda_s(j) = 1 - a_s(j)$, $\mu_s(j) = \sum_{t \neq s} \max(-a_t(j), 0)$, $\sigma_s(j) = \sum_{t \neq s} \max(a_t(j), 0)$, $E_s(j) = \max(\mu_s(j) - \lambda_s(j), 0)$, $\Phi_s(U) = \sum_j \max(\beta_s(j), 0) * E_s(j)$, and $S*_s(U) = \sum_j \max(\beta_s(j), 0) * (\sigma_s(j) + 2 * \max(-\lambda_s(j), 0))$; then for every pivot s , $S*_s(U) \leq 2 * \Phi_s(U) + 6 * \delta(P)$.*

Status. proved; af: validated. This is one of the rigorous in-repo results.

Proof tree. The registry shard records an 11-live-node adversarial tree (12 total nodes, 1 archived), root `validated`, taint 12/12 `clean`, closed in run 1 after five rounds. The export is `proofs/lem-factorization/e`. The contract was narrowed to the single inequality before seeding; the inherited tightness claim for the constants (2,6) was deliberately not elevated and remains `proved-mod-audit` material in the registry shard body.

Role in the chain. This is the composition link of the (EX) program: any universal chart bound $\max_s \Phi_s(U_0) \leq C_0 \delta(P)$ over the $\theta = 1/2$ class now composes rigorously to the class-score constant $C_{sf} = 2C_0 + 6$. The (EX) input itself (1) remains a conjecture: certified numerical evidence supports a plateau near $C_0 = 2$, but nothing in this section promotes it.

9 Status ledger for non-validated registry results

The rows below are anchors, not upgrades. They give every remaining registry result a report label while preserving the registry status. The seven **af**-validated results are reproduced in 1–7; they are not duplicated in this ledger.

Table 1: Registry status ledger for report anchors outside the seven validated sections.

Registry id	Status	Contract gloss
<code>conj-ex</code>	conjecture	(EX) chart-bound working form for exact signed idempotents; still a conjectural input equivalent to <code>conj-kernel</code> in the registry.
<code>conj-kernel</code>	conjecture	Kernel Conjecture, the geometric input intended to close <code>op-exposed-hull</code> and hence <code>op-classical</code> .
<code>conj-no-free-frontier</code>	conjecture	Exposedness-absorption statement from the frontier analysis; retained as conjectural and not elevated.
<code>ex-hume</code>	proved-mod-audit	Explicit 3-by-3 sharpness family for the exponent in <code>op-classical</code> ; inherited proof is not rigorous here.
<code>lem-canonical-separator</code>	proved-mod-audit	Canonical separator package for a hidden top vertex and its harmonic deficit; inherited/mod-audit only.
<code>lem-dual-localization</code>	open	Frame-free dual-localization gap behind the linear-law route; explicitly open.
<code>lem-exposed-circuit</code>	proved-mod-audit	Concentration and circuit estimate for exposed vertices; inherited/mod-audit only.
<code>lem-leakage</code>	proved-mod-audit	Affine-face leakage estimate for almost-idempotent stochastic rows; inherited/mod-audit only.
<code>lem-wiggle-rigidity</code>	proved-mod-audit	F-WR wiggle-rigidity estimate for common-pattern webs; side conditions remain inherited.
<code>obs-deep-leakage</code>	heuristic	Heuristic obstruction showing why a purely local row-identity argument cannot price deep-side mass.
<code>obs-fwr-gap</code>	heuristic	F-WR forbidden-gap obstruction to using web rigidity as the dimension-free shallow-class counter.
<code>obs-linear-law-finite-delta</code>	numerical	Exact finite-delta certificate exceeding the empirical linear-law constant; L3 evidence, not proof.
<code>obs-sigma-halo-nonrobust</code>	numerical	Exact certificate that the literal zero-halo sigma cap is false; L3 evidence and obstruction.
<code>op-classical</code>	open	North-star classical projection stability problem; still open in this repo.
<code>op-exposed-hull</code>	open	Global exposed-hull lemma that would imply <code>op-classical</code> through the inherited factorization route; open.
<code>prop-approx-simplex</code>	proved-mod-audit	Approximate-simplex reduction to a stochastic idempotent; inherited/mod-audit only.
<code>thm-classical-factorization</code>	proved-mod-audit	Conditional classical factorization theorem through cluster geometry and <code>op-exposed-hull</code> ; not rigorous here.
<code>thm-cluster</code>	proved-mod-audit	Cluster geometry reduction for signed affine retractions; inherited/mod-audit only.
<code>thm-corner-constants</code>	proved-mod-audit	Corner-scale constants, with numerical components explicitly not promoted.
<code>thm-rank-one</code>	proved-mod-audit	Rank-one signed affine retraction stability, including the Hume family; inherited/mod-audit only.

<code>thm-simplex</code>	proved-mod-audit	Simplex and one-dimensional row-polytope stability reduction; inherited/mod-audit only.
<code>thm-well-exposed</code>	proved-mod-audit	Well-exposed-vertex route into the simplex theorem; inherited/mod-audit only.
